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Assignment-9

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2023148

Q1)

```
# include <stdio.h>
# include <stdlib.h>
# include <stdlib.h>
```

~~int main() {~~

```
int main (int argc, char * argv[]){

        if (argc != 2){
            printf("Usage is %s <virtual-address>\n",
                                 argv[0]);

    return 1;
}

uint 32-t virtual-address = (uint32-t).strtoul
                    (argv[1], NULL, 10);

const uint32-t PAGE-SIZE = 4096;

uint 32-t page-number = virtual-address / PAGE-
                                          SIZE;

uint 32-t offset = virtual-address % PAGE-SIZE;

return 0;

}
```

```
gcc ques-1.c -o ques-1
PS C:\Users\USER\Desktop\OSLab\Assignment-9> .\ques-1.exe 19986
The address 19986 contains:
Page number = 4
Offset = 3602
PS C:\Users\USER\Desktop\OSLab\Assignment-9>
```

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Q2. Write a program that implements the FIFO and LRU page replacement algorithms. First, generate a random page reference string where page numbers range from 0 to 9. Apply the random page reference string to each algorithm, and record the number of page faults incurred by each algorithm.

```
# include <stdio.h>
# include <stdlib.h>
# include <stdbool.h>
# include <time.h>

# define MAX_PAGE_FRAMES 7
# define MIN_PAGE_FRAMES 1
# define MAX_PAGE_NUMBER 9
# define REFERENCE_STRING_LENGTH 100

void generate reference string ( int* reference string,
                                 int length ) {

    for (int i=0 ; i< length ; i++){
        reference string [i] = rand ()%
                         (MAX_PAGE _NUMBER+1)

page Exists (int *frame_Array , int numFrame,
                                int page){

    for (int i=0 ; i<NUMFRAME ; i++) {
        if (frameArray[i] == page) {
                       return true;
        }
    }
```

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```
int findOldestPartition(int *frameArray, int numFrames,
                        int *ages, int *currentTime){

    int oldest = 0;
    for (int i = 1 ; i < numFrames ; i++){
        if (ages[i] < ages[oldest]){
                oldest = i;
        }
    }
    return oldest;
}

int findLRUPosition( int *frameArray, int numframes, int *la
                                                       {

    int lruPos = 0;
    for (int i = 0 ; numFrames ; i++){
        if (lastused[i] < lastused[lrupos]){
                lrupos = i;
        }
    }
    return lrupos;
}

int fifo ( int *refurene.string , int length, int numFrames){

    int frame-Array [MAX-PAGE-FRAMES];
    int ages [MAX-PAGE-FRAMES];
    int page Faults = 0;
    int currentTime = 0;
```

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```
for (int i=0 ; i<numframe ; i++) {
        frame ; Array (i) = -1;
        ages (i) = -1;
}

for (int i=0 ; i<length ; i++) {
        int page = referenceString (i);
        if ( !pageExists (frame Array, numFrames,
                pagefaults ++;
                int emptyFramePos = -1;
            for (int j=0 ; i<numframes ; i++) {
                if (frameArray [j] == -1) {
                    emptyFramePos = -i;
                    break;
                }
            }
            if (emptyFramePos != -1) {
                frameArray [emptyFramePos] = page;
                ages [emptyFramePos] = currentTime + 7;
            }
    } else {

            int oldestpos = findoldest position (
                    frameArray, numFrame, ages,
                    currentTime;
            }
        }
    }
    return pagefaults
}
```

```
int lru ( int * reference string , int length, int numframes)
{
    int frameArray [MAX_PAGE_FRAMES];
    int lastUsed [MAX_PAGE_FRAMES];
    int PageFaults = 0;
    int currentTime = 0;
    for (int i = 0 ; i < numFrames ; i++){
            frameArray [i] = -1;
            lastused [i] = -1;
    }

int main() {
    srand ( time (NULL) ){
        int referencestring [REFERENCE_STRING_
                                LENGTH];
    generateReferenceString (referencestring,
                REFERENCE_STRING_LENGTH);

  printf ("Generated page reference string : \n");

  return 0;
}
```

```
Reference string: 3 7 4 6 6 2 8 8 4 0 3 5 2 2 2 6 3 2 2 2

For 1 frames:
FIFO Page Faults = 14
LRU Page Faults  = 14

For 2 frames:
FIFO Page Faults = 14
LRU Page Faults  = 14

For 3 frames:
FIFO Page Faults = 13
LRU Page Faults  = 13

For 4 frames:
FIFO Page Faults = 11
LRU Page Faults  = 11

For 5 frames:
FIFO Page Faults = 11
LRU Page Faults  = 11

For 6 frames:
FIFO Page Faults = 9
LRU Page Faults  = 10

For 7 frames:
FIFO Page Faults = 9
LRU Page Faults  = 8
```